Pilot 2 — Technical Protocol (Draft for External Review)

Working title: *Does an Agent's Own Change Keep Time? Testing an Experiential Index of Adaptation Against Clocks, Counts, and Logged Update Summaries*

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Status: draft protocol **v0.2**, 27 October 2026 (citations updated 1 November 2026; no change of design). Not frozen. To be frozen, hashed (SHA-256), and deposited on Zenodo before any main run. v0.2 revises v0.1 after an AI review (Google Argon) and an AI analysis of that review (Copilot); every change and every rejected suggestion is listed in §19.
Builds on: Ionici (2026), *A Time of Their Own* v1.3, §4 and Level 2 of Table 2, https://doi.org/10.5281/zenodo.23847648; method of Ionici (2026), *Does Explicit Episode–Trace Binding Help?* v1.3, https://doi.org/10.5281/zenodo.23847602; code base https://doi.org/10.5281/zenodo.23847723.

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0. Position within the research programme (context for the reviewer)

This protocol is the empirical continuation of two published works by the same author. Neither is under review here. They are summarised so that every design decision below can be traced to its source.

0.1 The Perspective: what is to be tested

A Time of Their Own (v1.3) argues the following:

- **The problem.** A persistent artificial agent whose complete state can be saved and restored holds no evidence about elapsed time or about its own change that it could not have been given, reset, or forged together with that state.
- **The classification.** It classifies the temporal evidence available to an agent (order, duration, sequential computation, continuity, and its own change) by who can write, reset, forge, or copy it, and by whether the agent's self-model can read it.
- **Two roles that must be kept apart:**
    1. A **continuity anchor**: a non-restorable time reference c_a and a state-bound operation counter N, read through a trusted path. It detects and distinguishes suspension and restoration (Level 4 of its programme).
    2. A **dual clock**: an **experiential index** $c_e(t) = \Sigma_{\tau<t} D(S(\tau), S(\tau+1))$, accumulated change of the agent's persistent state S, read against c_a. It serves the agent's model of its own adaptation (Level 2). The index is non-decreasing, is updated online at constant cost, and is distinct from the net displacement $D(S(t_0), S(t_1))$. The choice of D is declared an open problem.
- **Admissibility of D.** A candidate D is admissible only if it:
    - (a) beats decoders restricted to within-episode (local) update statistics;
    - (b) survives the reset criterion and randomisation of episode order, where it is meant to reflect history;
    - (c) predicts a later functional consequence better than event counts and logged update summaries;
    - (d) is used by the agent rather than merely decodable from it.
- **Test programme.** Six levels (time reference, experiential index, history, discontinuity, closed-loop use, temporal self), each paired with the conventional competitor that can close it. A level is **closed** if its competitor is non-inferior on the level's primary metric within a margin fixed in advance and no worse on any declared cost dimension; a mixed trade-off is reported as unresolved, neither closing the level nor supporting the hypothesis. The Perspective states that each level needs its own evaluation contract; this

protocol is that contract for Level 2. The primary metric of Level 2 is **predictive validity per unit of storage**.
- **Stance on outcomes.** Equivalence with the competitor is a legitimate negative result, and *functional separation* (the platform suffices, and agent-held evidence is unnecessary) is a possible outcome of the programme. The Perspective offers its tests to others; versions 1.0–1.2 also stated that its author did not plan to pursue them. This study is a declared change of that plan.

This protocol tests Level 2 only: whether c_e, under some admissible D, is worth holding.

0.2 The closed study: how it is to be tested

Does Explicit Episode–Trace Binding Help? (v1.3, closed) asked whether binding an episode record to a summary of the internal change it produced (arm D) helps a model attribute the episode's source (own, imported, or inferred), compared with an information-matched control that receives the same streams without the binding (arm F). The model was a small Transformer self-model reading the traces of a plastic agent. Over five exploratory iterations:

- **The apparent advantage was a control failure.** Generic controls failed to use the trace, which produced a large but uninformative advantage for D (+0.38 Brier skill score). When the control was made competent (87% of a decoding ceiling), D − F was +0.002 BSS, with descriptive equivalence under a ±0.05 BSS margin.
- **The signal was local, not historical.** Decoder diagnostics showed that the source was carried by **local update statistics**, and that resetting the state before each episode did not reduce decodability. The benchmark therefore did not test attribution from path-dependent history.
- **A shortcut caused the apparent reversal.** With 32 episodes per window, an apparent reversal between objectives was traced to a shortcut in how one objective was applied, and removing it repaired the reversal.

What carries over is the method, not a result: an information-matched control, a competence audit of the control, a decoding ceiling, a reset criterion, equivalence tests (TOST) with a prespecified margin, the distinction between onset (whether learning starts) and skill (performance once learned), reliance versus necessity, frozen and hashed configurations, and a per-run inventory.

What carries over as a warning: local update statistics can impersonate accumulated history, and an under-specified control can manufacture an advantage. This is why competitor C6 is mandatory and why the baseline is deliberately strong (§7).

0.3 Why Level 2, and why not the continuity anchor

An earlier internal draft of Pilot 2 tested the continuity anchor (Level 4) in simulation. It was abandoned. With a simulated anchor that is deterministic by construction, classifying suspension against restoration reduces to comparing two counters. A raw anchor can then beat an information-matched adjudicated flag only where the flag is absent, which is an argument about availability, not an empirical finding.

Level 2 is different: whether c_e predicts the agent's own forgetting depends on the actual learning dynamics of the network, so the outcome is not fixed by the design.

Prior work with the same index. Feng et al. (2026, FOREVER, ACL 2026, arXiv:2601.03938) define a "model-centric time" as the cumulative L2 norm of parameter updates, $\tau_t = \Sigma_{i\leq t} \|\Theta_i - \Theta_{i-1}\|_2$ on LoRA weights. This is c_e^{D1} (their τ is reset at each new task). They use it to schedule replay at forgetting-curve intervals, with models from 0.6B to 13B parameters. Their only ablation of the time signal is against step counts; learning-rate-based and gradient-norm-based signals are not tested. Priority for the use of accumulated update

magnitude as a model clock for replay therefore belongs to FOREVER, and this study claims no novelty for the index itself.

What this study adds is an audit of identification: whether such an index carries information beyond the union of cheap online competitors that could reconstruct it (§7), under a placebo that destroys its temporal alignment (G4) and under an adversarial stream with much change and little interference (F6). If the answer is equivalence, the benefit attributed to a model-centric clock is reproducible from elementary optimizer statistics; if it is superiority, the clock carries information of its own. Both outcomes bear directly on that line of work. Forgetting prediction more generally also exists (§18).

0.4 Content, not provenance

In the Perspective, the difference between agent-held evidence and logged update summaries is partly one of **provenance**: who writes, holds, and can reset or forge the evidence. In a simulation every quantity is computed by the same code, so provenance cannot differ between predictors. **Stage A therefore tests content only**: whether the number c_e carries predictive information that the other cheap numbers do not. For parameter-space distances (D1, D1_rel, D2), c_e is itself a cumulative update summary; H1 then asks whether *this* summary beats the strongest set of *other* cheap summaries. Claims about provenance are outside the scope of this study.

0.5 Traceability of design decisions

Design decision	Source
Definition of c_e; distinction from displacement; D as open problem	Perspective §4
Q1–Q4 and the mapping to criteria (a)–(d)	Perspective §4 (admissibility)
Confirmatory baseline B_strong; descriptive B_standard	Perspective §5 (a level closes if the competitor matches); closed study (control competence); review v0.1 (§19)
Competitor C6 (local update statistics)	Closed study: main diagnostic finding
Primary metric: validity per unit of storage (H2)	Perspective §5, Level 2
Three-way outcomes (supported, equivalent, inconclusive) and TOST	Perspective (equivalence as a legitimate result); closed study (equivalence margins)
Ceiling G1, competence G3, onset versus skill G6, reliance G8	Closed study, method; Perspective §6
Frozen configurations, hashes, inventory, bit-identical reruns G7	Closed study and its code archive
Competitors C1, C5b–C5d, C8, C8b, C9	External literature on forgetting and consolidation (§18)
Stage C on pretrained language models	Perspective (claims concern persistent agents, not toy worlds)

0.6 What the reviewer is asked to evaluate

Only this protocol: whether its design can deliver an interpretable answer to Q1–Q4 at the stated scale. The claims of the Perspective and the results of the closed study are published and are not under review.

0.7 Glossary

Term	Meaning
S, θ	Persistent state of the agent; its parameters
m, v; $\hat{m}$, $\hat{v}$	Adam first and second moments; bias-corrected values

Term	Meaning
c_a	Elapsed time in ticks (§4.5)
N, N(t)	Committed optimizer steps; their count at time t
c_e^D	Experiential index: accumulated D-change of S
$\bar{\rho}$	Mean interval rate Δc_e / ΔN; the only information in c_e beyond counts (§2)
Disp^D	Net displacement $D(S(t_0), S(t_1))$
√JS	Jensen–Shannon distance, base-2 logarithm, in [0, 1] (a metric)
m_ch	Chunk stride, in optimizer steps, for chunked distances (§5)
L_reh	Length of one rehearsal segment (§9)
B_standard	Descriptive baseline: standard telemetry (§7)
B_strong	Confirmatory baseline: all cheap competitors (§7)
SESOI	Smallest effect size of interest
TOST	Two one-sided tests (equivalence)
MET	Minimal-effect test: lower confidence bound above the SESOI
BSS	Brier skill score (used in the closed study)
Gate	A prespecified validity check; failing it changes how a result may be reported

1. Research question

Let a Transformer learn continually from a non-stationary stream of tasks. Let

$c_e(t) = \Sigma_{\tau<t} D(S(\tau), S(\tau+1))$

be its *experiential index*: the accumulated change of its persistent state S under a distance D. It is a non-decreasing scalar held by the agent.

Storage. Maintaining c_e costs **one running scalar**. Answering a query about task k needs the snapshot $c_e(t_0,k)$ taken at its last exposure: **one scalar per tracked task exposure**, at most K + 1 scalars for K active tasks (here $K \leq 40$). This is O(1) per update and O(K) for queries, against a full snapshot per task for displacement (§5.2).

Q1 (predictive validity). Does Δc_e over an interval $(t_0, t_1]$ predict the agent's forgetting of a task last trained at t_0 better than a strong set of cheap online competitors: elapsed time; step, token, and event counts; cumulative learning rate and loss; logged gradient summaries, including learning-rate-weighted ones; optimizer and parameter-state norms; local (within-window) update statistics; and the loss change on a fixed reference battery?

Q2 (storage efficiency). How much of the predictive value of the full net displacement $D(S(t_0), S(t_1))$, which requires storing a snapshot at t_0, does the scalar c_e retain?

Q3 (path dependence; secondary). Under a fixed multiset of experiences, do order-induced differences in forgetting track order-induced differences in Δc_e?

Q4 (use). Does a rehearsal scheduler that uses c_e outperform budget-matched schedulers that use the competitors, and does its advantage depend on the actual c_e stream rather than on a placebo stream?

The admissibility criteria for D stated in *A Time of Their Own* §4 map onto these questions:

Criterion	Question
(a) beats decoders restricted to local update statistics	Q1, competitor C6, gate G5
(b) order sensitivity where history matters	Q3 (secondary, §10.3)
(c) predicts a functional consequence better than counts and logged summaries	Q1
(d) used by the agent rather than merely decodable	Q4, gate G8

Scope restriction on criterion (b). The reset criterion is **not applicable** to the target used here: resetting $S(t_0)$ would erase the task whose forgetting is measured. Only order sensitivity is examined, and only as a secondary hypothesis. The rate of change $\rho(\tau)$ still depends on the whole prior history through $S(t_0)$; this is why absolute run progress $N(t_0)$ and state norms at t_0 are covariates (§7).

2. Identification note (read first)

Within a run, at a fixed evaluation time t_1, the ranking of tasks by $\Delta c_e = c_e(t_1) - c_e(t_0,k)$ is identical to their ranking by $t_1 - t_0,k$, the time elapsed since their last exposure (ties occur where $D = 0$, for example during idle intervals). The reason is that c_e is non-decreasing. A global scalar index therefore carries **no task-specific ordinal information at a fixed time**.

It does carry **cardinal** information: the spacing between two tasks' values, $c_e(t_0,k) - c_e(t_0,j)$, equals the change accumulated between their exposures, which varies with the rate $\rho(\tau) = dc_e/dN$. All information that c_e carries beyond counts lies in this **temporal variation of its rate**: some intervals of equal length contain more change than others. Three consequences follow:

1. Q1 is identifiable only to the extent that the interval rate $\bar{\rho} = \Delta c_e / \Delta N$ varies across intervals and is not reconstructible from the baseline. Gate G2 checks this on the rate, not on the raw increment, because the raw increment is dominated by interval length.
2. A scheduler that *selects which task to rehearse* by ranking tasks on Δc_e since their last exposure is equivalent to least-recently-rehearsed. Q4 must therefore test *when*, and *how much*, to rehearse (allocation over time), or use c_e together with task covariates inside a learned predictor (§9).
3. c_e is not task-specific memory. It cannot by itself encode which tasks a stretch of change interfered with (§10.3).

3. Agent

Component	Specification
Architecture	Decoder-only Transformer, 4 layers, d_model = 128, 4 heads, d_ff = 512, pre-LN, GELU, learned positional embeddings, maximum context 64, vocabulary 512 tokens. Input embedding and output head **untied**; biases in all linear layers and LayerNorms. ≈ 0.93 M parameters (exact count recorded at Stage 0)
Sequences	One example per sequence, no packing; sequence length equals example length (4–6 tokens); padding to the batch maximum with masked loss
Optimizer (primary)	AdamW, $\beta = (0.9, 0.999)$, $\varepsilon = 1e\text{-}8$, weight decay $\lambda = 0.01$, global gradient clipping at norm 1.0 applied immediately before `optimizer.step()`. m and v see the clipped gradient $\tilde{g}$
Optimizer (robustness arm)	SGD with momentum 0.9, same clipping, learning rate and weight decay frozen at Stage 0. A passive EMA $v_\tau = 0.999\, v_\{\tau-1\} + 0.001\, \tilde{g}_\tau^2$ is maintained so that D2 is defined; it does not affect the update
Persistent state S	Parameters θ and optimizer moments (m, v); distances are computed on θ

Component	Specification
Logged per step	$\|g_\tau\|_2$ (raw) and $\|\tilde{g}_\tau\|_2$ (clipped); η_τ; batch size B_τ; training loss L_τ; $\|\Delta\theta_\tau\|_2$ before and after the decoupled weight-decay term; $\|\hat{m}_\tau\|_2$
Precision, device, determinism	float32 on CPU; `torch.use_deterministic_algorithms(True)`; fixed seeds for data, initialisation, and dropout (dropout = 0 in the primary arm). Each run is a **single-threaded worker** (`torch.set_num_threads(1)`, `torch.set_num_interop_threads(1)`, `OMP_NUM_THREADS=1`, `MKL_NUM_THREADS=1`); eight workers run in parallel
Software	Python 3.14.6, PyTorch 2.13.0 (CPU build), NumPy 2.5.1, scikit-learn 1.9.0, as installed at drafting; exact versions are re-recorded in `environment.txt` at freeze

4. World and streams

4.1 Task types and vocabulary

- **Fact tasks.** Task k is a set of 256 key→value pairs. The input is `[TASK_k] [KEY] →`, and the target is a single `[VALUE]` token. Keys are drawn from a target key pool P_{target}, and the overlap between tasks is controlled by a pool-sharing parameter $\pi_{overlap}$. Values are drawn uniformly from V value tokens. Fact probes are **trained items**: they measure in-sample associative retention.
- **Rule tasks.** Task k is an algorithmic map $y = (a_k \cdot x + b_k) \bmod m_k$, with $m_k \leq 64$ so that y is a single value token. The input is `[TASK_k] [D1] [D2] →`, where $x \in [0, 1023]$ is written as two base-32 digit tokens. Probe inputs are held out from training: rule probes measure held-out algorithmic generalisation.
- **Reference tasks.** A fixed set of fact and rule tasks, trained only during the childhood phase. They are never targets. They supply the reference battery R (§5) and the behavioural battery B_{beh}. Their keys come from a pool P_{ref} disjoint from P_{target}, and their rule moduli are not used by any target task.
- **Vocabulary partition** (frozen at Stage 0; indicative): 64 task tokens, P_{target} and P_{ref} key tokens (with multi-token keys if more keys are needed), 64 value tokens, 32 digit tokens, and special tokens, in total ≤ 512.
- **Exposure.** Every training, re-exposure, and rehearsal segment iterates over balanced permutations of the task's items (full epochs), so that every item of halves A and B has the same exposure count and comparable recency at the end of a segment.

4.2 Probes

Each target task has a probe set: its 256 facts (fact tasks) or 256 held-out inputs (rule tasks). The probe set is split **once, at task creation**, into disjoint halves A and B of 128 items each.

- **Half A** supplies the confirmatory target T1.
- **Half B** supplies the competence covariates at t_0, the inclusion criterion, the reliability ceiling (with A), and the Stage B retention target T3.

No probe item belongs to R or to B_{beh}. All probe losses and probabilities are read at the single answer position after →; cross-entropy is in nats.

4.3 Run structure

Phase	Steps	Content
Childhood	exactly 3,000	Reference tasks. Gate: reference accuracy ≥ 0.95 at the end; a run that fails is replaced and counted
Life	≈ 20,000	≈ 40 target tasks, presented according to the stream family; rehearsal segments, re-exposure segments, and idle intervals according to the family

Phase	Steps	Content
Evaluation	—	Probes of every learned task are evaluated at the end of every segment and on a grid every 200 optimizer steps

All segment lengths and the evaluation grid are integer multiples of 50 optimizer steps, so that chunk boundaries (§5) never fall inside a segment.

4.4 Stream families

Family	Generator	Intended decoupling	Role
F1, natural	New task every fixed number of steps, fixed segment length, round-robin rehearsal, constant LR, fixed batch	None	Positive control: counts are expected to suffice
F2, variable novelty	New tasks interleaved with re-exposure to already-mastered data in random amounts; batch size ∈ {16, 32, 64} per segment	Steps and tokens decouple from change	Pooled
F3, LR schedule	Cosine annealing with warm restarts at random times	Steps decouple from change; LR-weighted summaries are strong competitors	Pooled
F4, idle	Random idle intervals with no updates; c_a advances, θ is frozen	Only elapsed time decouples (C1–C7 stop with c_e)	Sanity check only; **not pooled** into H1
F5, held-out	Mixture of the F2–F3 generators with a different proportion of rule tasks, a different π_overlap, and idle intervals	Generalisation across generator parameters	Never used for fitting or selection; test only; reported alone
F6, adversarial	Target exposures at a regular rate, interleaved with variable-length distractor segments on a disjoint distractor pool ($P_{distractor} \cap P_{target} = \emptyset$) with label noise 0.3 and batch size 16, under the same optimizer configuration	High parameter change with little interference: favours the competitors	Never used for fitting or selection; test only; reported alone
O, order blocks	Designed blocks (§10.3) forked from one saved state	Path dependence	H3 only; **not pooled** into H1, H2, H4

F5 is built from the F2–F3 generators, so it tests generalisation across generator parameters, not freedom from designed decoupling. F6 tests the converse regime that F2–F5 lack: much movement, little forgetting. Scale-drift interventions (weight-decay pulses, rescaling of pre-LN weights) are exploratory robustness checks only, because they have no counterpart in ordinary training. Generator parameters for each family are frozen with the protocol.

4.5 Time

c_a counts *ticks*. Each optimizer step adds 1 tick; each idle interval adds its length in ticks; evaluations add 0 ticks. Without idle intervals c_a equals N. All interval sums in §7 run over committed optimizer steps $\tau \in (N(t_0), N(t_1)]$, so idle ticks contribute nothing to them.

5. Distances D and the index c_e

5.1 Candidate distances

D	Definition	Evaluated
D1	$\|\|\theta(\tau+1) - \theta(\tau)\|\|_2$	Every step
D1_rel	$(\Sigma_\ell \|\|\Delta\theta^{(\ell)}(\tau)\|\|_2^2 / (\|\|\theta^{(\ell)}(\tau)\|\|_2^2 + 10^{-12}))^{1/2}$, layer-wise, with Δθ taken **before** the decoupled weight-decay term	Every step
D2	$(\Sigma_i \hat{v}_i(s) (\Delta\theta_i(\tau))^2)^{1/2}$, with $\hat{v}(s)$ the bias-corrected second moment **frozen at the start s of the current segment**	Every step
D3	Mean over $x \in R$ of $\sqrt{JS}(p_{S(\tau)}(\cdot\|x), p_{S(\tau+m_{ch})}(\cdot\|x))$ at the answer position, $\|R\| = 256$	Every m_ch steps (chunked)
D4 (secondary)	√(1 − linear CKA) of final-layer answer-position representations on R (Kornblith et al. 2019)	Every m_ch steps (chunked)

Why D2 uses a lagged curvature. Under Adam, $\Delta\theta_i \approx -\eta\, \hat{m}_i / \sqrt{\hat{v}_i}$. Weighting by the *current* $\hat{v}$ therefore cancels the preconditioner: $\Sigma_i \hat{v}_i(\tau)\, \Delta\theta_i(\tau)^2 \approx \eta_\tau^2\, \|\hat{m}_\tau\|_2^2$, so a same-step D2 would be the learning-rate-weighted momentum norm, a logged optimizer summary. The segment-lagged weighting avoids exact cancellation; whatever remains of the relation is controlled by competitor C5d (§7).

Why D1_rel. In a pre-LN Transformer, some weight rescalings change $\|\theta\|$ without changing the function, and AdamW shrinks θ at every step independently of the data. D1_rel removes both scale effects; raw D1 is kept for comparison.

- $c_e^D(t)$ = sum over steps (D1, D1_rel, D2) or over chunks (D3, D4) up to t. Chunks are indexed by committed optimizer steps.
- √JS is a metric, so the chunked D3 sum is a lower bound on the per-step sum. √(1 − CKA) is the chordal distance between centred Gram matrices and also satisfies the triangle inequality.
- **Chunking control.** On calibration runs, chunked versions of D1, D1_rel and D2 at the same stride m_ch are computed alongside the per-step versions, so that a difference between D3 and D1/D2 cannot be produced by temporal smoothing alone. Sensitivity: m_ch ∈ {1, 5, 10, 20, 50} on 5 calibration runs (m_ch = 1 over a 2,000-step window). m_ch = 10 is adopted if, for D3, the within-run Spearman correlation between Δc_e at m_ch = 1 and m_ch = 10 across intervals is ≥ 0.98 and their ΔR^2 differ by ≤ 0.01; otherwise m_ch = 5 (with |R| reduced to 128 if needed for cost).
- **Stationarity check of R.** Because R is never retrained after childhood, the sensitivity of D3 may decay as R is forgotten. On F1 calibration runs, the mean D3 rate in the fourth quartile of the life phase must be within ±20% of the first quartile. If not, a fixed maintenance rehearsal of R (one batch every 500 steps, in all families) is added before freeze.
- **D*** is the single D used in the confirmatory tests. It is chosen among D1, D1_rel, D2 and D3 on calibration runs as the one with the largest incremental validity over B_strong (§7), and is frozen before any test run.

5.2 Net displacement and its storage

$Disp^D(t_0, t_1) = D(S(t_0), S(t_1))$, evaluated at grid points. The cached object per active task exposure is:

D	Cached at t_0	Numbers per t_0
D1, D1_rel	$\theta(t_0)$ (and layer norms)	≈ 0.93 M
D2	$\theta(t_0)$ and $\hat{v}(t_0)$ (Fisher weight anchored at t_0, as in EWC)	≈ 1.9 M
D3	answer-position output distributions on R	256 × 512 ≈ 0.13 M
D4	final-layer answer-position representations on R	256 × 128 ≈ 0.03 M

A snapshot of task k is overwritten at its next exposure, so at most $K \leq 40$ snapshots are active. Storage ratios in H2 are reported per D (of order 10^{-5} to 10^{-6} in Stage A).

6. Targets

T1 (primary): forgetting. For each target task k and each exposure segment of k (training, re-exposure, or rehearsal) ending at step s_0, let t_0 be the evaluation immediately after s_0. For every evaluation time $t_1 > t_0$ strictly before the first step of the next segment that trains k:

$$F_k(t_0, t_1) = \bar{L}_{A,k}(t_1) - \bar{L}_{A,k}(t_0),$$

where $\bar{L}_{A,k}$ is the mean cross-entropy on probe half A of task k.

- **Inclusion:** a triplet is included only if the probe-**B** accuracy of task k at t_0 is ≥ 0.90, and $\Delta N = N(t_1) - N(t_0) \geq 50$ (or $\Delta c_a \geq 50$ ticks where no step is taken). Excluded triplets are counted and reported by reason.
- **Unit of observation:** the triplet (k, t_0, t_1). **Unit of inference:** the run.
- **Weighting:** each exposure episode (k, t_0) carries equal total weight within its run: $w = 1 / (E_r \cdot M_{r,k,t_0})$, where E_r is the number of valid episodes in run r and M_{r,k,t_0} the number of evaluation times of that episode. The same weights are used in model fitting (`sample_weight`) and in R^2_r.

T1-acc (secondary): bounded forgetting. $\Delta Acc_{A,k} = Acc_{A,k}(t_0) - Acc_{A,k}(t_1)$ on the same triplets. It checks that an effect on T1 reflects functional forgetting rather than logit-scale drift. The H1 statistic is reported for T1-acc descriptively.

T2 (secondary): behavioural drift. The $\sqrt{JS}$ distance between the outputs of $S(t_0)$ and $S(t_1)$ on B_{beh}. It is reported descriptively. T2 is structurally favourable to Disp^{D3}, so it is not used for confirmatory claims.

T3 (Stage B): retention. The time-averaged probe-B accuracy over all learned tasks during the life phase, at a fixed rehearsal budget.

7. Predictor sets and competitors

All interval quantities are sums over $\tau \in (N(t_0), N(t_1)]$. Every competitor is agent-available online and costs ≤ 2 stored scalars per t_0.

Covariates X_cov, given to every model (agent-observable only):

- task type (fact or rule);
- $\bar{L}_{B,k}(t_0)$ and $Acc_{B,k}(t_0)$ (competence at t_0, from half B);
- $\bar{L}_{B,k}$ at the start of the exposure segment ending at s_0 (prior mastery, a proxy for novelty);
- input-token overlap of task k with the agent's history: $1 - \sqrt{JS}(q_k, q_{hist}(t_0))$, where q_k is the task's input-token unigram distribution and $q_{hist}(t_0)$ the cumulative input-token histogram up to t_0 (one 512-length counter);
- $N(t_0)$, absolute run progress; $N(t_0) - N(t_{first,k})$, task tenure; number of prior exposures of k; number of active learned tasks $K_{active}(t_0)$;
- $\|\theta(t_0)\|_2$ and $\sqrt{\mathrm{Tr}\,\hat{v}(t_0)}$.

The generator parameter $\pi_{overlap}$ is **not** a covariate: it is world metadata (see C8).

B_standard (standard telemetry; descriptive only):

Code	Predictor over $(t_0, t_1]$	Source
C0	Δc_a, elapsed ticks	wall clock
C1	ΔN, with the fixed basis $\{\Delta N, \log(\Delta N + 1), \log(\Delta N + 10), \log(\Delta N + 100)\}$	shifted power law of Kalajdzievski (2024), arXiv:2401.05605
C2	Δ tokens	data volume
C3	$\Sigma \eta_\tau$	cumulative learning rate
C4	ΣL_τ	cumulative training loss
C5	$\Sigma \|\|g_\tau\|\|_2$ (raw)	logged gradient norm
C7a	number of segment boundaries and of agent-triggered rehearsals	agent-observable event counts

B_strong (confirmatory baseline) = X_cov ∪ B_standard ∪:

Code	Predictor over $(t_0, t_1]$	Source
C5a	$\Sigma \|\|\tilde{g}_\tau\|\|_2$ (clipped)	gradient actually used by the optimizer
C5b	$\Sigma \eta_\tau \|\|g_\tau\|\|_2$, $\Sigma \eta_\tau \|\|\tilde{g}_\tau\|\|_2$, $\Sigma \eta_\tau L_\tau$	within-step products that interval sums cannot recover
C5c	$\Sigma \eta_\tau^2 \|\|\tilde{g}_\tau\|\|_2^2$, $\Sigma \|\langle \tilde{g}_\tau, \Delta\theta_\tau \rangle\|$	squared-gradient accumulator; path-integral work of Synaptic Intelligence (Zenke et al. 2017)
C5d	$\Sigma \eta_\tau \|\|\hat{m}_\tau\|\|_2$	learning-rate-weighted momentum norm (the same-step D2 identity, §5.1)
C2b	$\Sigma \eta_\tau / B_\tau$	noise-scale-weighted steps
C6	mean and SD of $\|\|\Delta\theta_\tau\|\|_2$ over the first W_eff and the last W_eff steps of $(t_0, t_1]$, with $W_{eff} = \min(20, \lfloor \Delta N/4 \rfloor)$, windows clamped inside the interval	local update statistics (closed study; criterion (a))
C6b	$\Delta\|\|\theta\|\|_2$ and $\Delta\sqrt{\mathrm{Tr}\,\hat{v}}$ over the interval	state-scale drift
C10	$L_R(t_1) - L_R(t_0)$: change of the mean loss on the reference battery R	canary loss; 1 stored scalar per t_0
rates	ΔN-normalised versions of C2–C5d and C10 ($Z / \max(\Delta N, 1)$)	computed, not stored

Why two baselines. B_strong is deliberately close to a reconstruction of c_e from cheap telemetry, so that a positive H1 cannot be explained by an omitted product of learning rate and gradient, by momentum, by weight decay, by norm drift, or by local statistics. Claimed advantages over under-trained baselines often vanish when the baseline is trained properly (Mukhoti et al. 2018); the closed study documented the same failure for an information-matched control. B_standard is reported so that readers can see how much of any gap is closed by the specialised competitors. If c_e beats B_standard but not B_strong, the result is reported as *reconstructible from cheap telemetry*, not as support.

Reference predictors (not agent-available, or of high storage; used to bound, not to compete; exploratory):

Code	Predictor	Storage
C7b	separate counts of new-task and re-exposure segments (oracle labels)	world metadata
C8	Σ over intervening tasks j of key overlap(k, j) × steps on j; for rule tasks, overlap is the Jaccard overlap of input-token sets	world metadata (Lee, Goldt and Saxe 2021, arXiv:2107.04384)

Code	Predictor	Storage
C8b	Disp^{D2}(t_0, t_1) with $\hat{v}(t_0)$, Fisher-weighted displacement (EWC: Kirkpatrick et al. 2017)	≈ 1.9 M per t_0
C9	First-order response $\langle \nabla_\theta L_{train,k}(S(t_0)), \theta(t_1) - \theta(t_0) \rangle$, with the gradient computed on task k's **training items only**, never on probe half A (cf. Su and Zhang 2026; Jin and Ren 2024)	≈ 1.9 M per task
Ceiling	Reliability of the half-A target: within-run correlation r_AB between F_A and F_B on the same triplets, **uncorrected** (the target uses one half)	—

8. Prediction models and competence audit

- **Primary model family:** gradient-boosted regression trees (scikit-learn `HistGradientBoostingRegressor`), `max_features = 1.0`, `max_bins = 255`. Monotone constraints (`monotonic_cst`): +1 for C0, C1 (raw ΔN only), C2, C2b, C3, C4, C5, C5a–C5d, Δc_e, Disp, C8, C8b; 0 for C7a, C6, C6b, C10, all rates, all of X_cov, and the log terms of C1.
- **Secondary model family:** ridge regression on the predictors with log1p applied to raw (non-log) positive interval sums, sign(x)·log(1 + |x|) for signed predictors (C6b, C10, C9), interactions with all of X_cov, a `StandardScaler` fitted on training runs, and λ chosen by cross-validation over training runs.
- **Equal tuning budget:** every predictor set receives the same random search, 50 configurations, on validation runs, with the same seeds. The best configuration of B_strong is always included as a candidate for B_strong + c_e. The **locked-hyperparameter** ΔR^2 (both sets fitted with the B_strong configuration) is the confirmatory statistic; the re-tuned ΔR^2 is reported alongside.
- **Fitting:** models are fitted on pooled, weighted triplets from the training runs and evaluated on test runs. No test-run information is used in fitting, selection, or transformation. Training-split variability is assessed by 5-fold cross-fitting over training runs (the five fitted models are averaged); a full two-sample bootstrap (resampling training and test runs, with refitting) is run for the ridge model as a sensitivity analysis.
- **Skill per test run r:** weighted $R^2_r = 1 - SSE_r / SST_r$, with SST_r around the weighted mean of run r. Weighted Spearman $\rho_{S,r}$ and MAE_r are reported alongside.
- **Decomposition (reported, not confirmatory):** within-episode skill, with SST computed around the mean of each episode (k, t_0) across its t_1; and partial skill [SSE_r(B_strong) − SSE_r(B_strong + c_e)] / SSE_r(X_cov).

9. Stage B: use by the agent

Stage B is run only if H1 is supported (§10.1).

- **Budget.** The rehearsal budget is fixed per run: K_reh = Q / L_reh rehearsal segments of L_reh = 50 steps over the life phase. The budget is a **hard cap**, and every scheduler spends exactly Q on every run (asserted per run; gate G7).
- **Pacing.** Threshold schedulers use an online pacing rule: $\theta_{eff}(N) = \theta_{cal} \cdot [(K_{reh} - k_{spent}(N)) / K_{reh}] / \max(\varepsilon, (N_{life} - N) / N_{life})$, with θ_{cal} calibrated on validation runs. When the remaining life steps fall to the remaining budget, all remaining rehearsals are executed (terminal flush). The same pacing rule is applied to every threshold scheduler.
- **Schedulers.** By the identification note (§2), schedulers differ in **when** they spend the budget, and in which task only through a learned predictor that includes covariates.

Scheduler	Rule
S-uniform	One rehearsal segment every ⌊N_life / K_reh⌋ optimizer steps; task chosen least recently rehearsed
S-c_e	Rehearsal triggered when Δc_e^{D*} since the last rehearsal ≥ θ_eff; task least recently rehearsed
S-C2, S-C3, S-C4, S-C5, S-C5b, S-C7a	As S-c_e, with the trigger on tokens, cumulative LR, cumulative loss, gradient norm, LR-weighted clipped gradient norm, or event counts
S-pred(B_strong)	A learned forgetting predictor (Stage A model on B_strong) scores every task; rehearse the argmax when the predicted maximum ≥ θ_eff
S-pred(B_strong + c_e)	Same, with Δc_e^{D*} added (information-matched pair)
S-oracle	Rehearse the task with the largest true current forgetting on half B (upper bound)
S-random	Random times and tasks, same budget

- **Relation to FOREVER.** With D* = D1, S-c_e is a trigger on the clock of Feng et al. (2026). An exploratory arm S-FOREVER applies their schedule (replay when c_e^{D1} since the end of a task's training reaches $d \cdot \tau_day$, with τ_day the change over the first 24 steps and $d \in \{1, 2, 4, 7, 15, 30, \ldots\}$), under the same budget cap and pacing.
- **Comparator selection.** The single best trigger competitor S-C* is chosen among S-uniform, S-C2, S-C3, S-C4, S-C5, S-C5b and S-C7a on **validation runs**, frozen, and evaluated once on test runs.
- **Online covariates.** In Stage B, competence covariates at t_0 are computed from the loss and accuracy on the last training mini-batches of the segment, not from probes. T3 uses half B only, and no scheduler except S-oracle reads any probe.
- **Placebo for reliance (G8).** Information that is decodable need not be used (Elazar et al. 2021); reliance is therefore tested by intervention. S-c_e and S-pred(B_strong + c_e) are rerun with the c_e stream replaced by a within-run block-permuted stream (§11, G4); a cross-seed yoked stream from the same family, aligned by N, is run as a robustness check.
- **Logged:** realised rehearsal steps per run; Wasserstein distance between the trigger times of S-c_e and of S-C* (when); task agreement at matched times (which).

10. Hypotheses

10.1 Decision rules

- **One-sided level:** α = 0.025 per side for every test. Each hypothesis is decided from a single two-sided (1 − 2α_adj) BCa bootstrap confidence interval (10,000 resamples over the resampling unit), where α_adj is the level assigned to it by the procedure below:
  - **supported** if the lower bound is > δ (minimal-effect test);
  - **equivalent** if the whole interval lies within (−δ, δ) (TOST, each side at α_adj; Lakens et al. 2018; for stochastic machine-learning experiments, Knopov and Shvai 2026);
  - **inconclusive** otherwise.
- **Multiplicity (fixed sequence, then Holm).** H1 is tested first at α = 0.025. Only if H1 is supported are H2, H4a and H4b tested, as a Holm family at α = 0.025: ranked by their one-sided bootstrap p-values, the j-th is tested at $\alpha_adj = 0.025 / (3 - j + 1)$. This controls the family-wise error rate. If H1 is equivalent or inconclusive, H2–H4 are not tested confirmatorily and Stage B is not run.
- **SESOI (rule frozen now, value computed on calibration runs and frozen before test):** $\delta = \max(0.02, \min(0.05, 0.15 \times [\text{ceiling} - \bar{R}^2_cal(B_strong)]))$, where the ceiling is $\bar{r}_AB$ from G1. Raw ΔR^2 and ΔR^2 as a fraction of the headroom are both reported. Stage B: δ_B = 2 percentage points of T3.

10.2 Confirmatory hypotheses

	Statement	Statistic	Resampling unit
H1	Adding Δc_e^{D*} to B_strong increases held-out skill on T1	Mean over test runs of the locked-hyperparameter ΔR^2_r = R^2_r(B_strong + Δc_e^{D*}) − R^2_r(B_strong); test runs from F1, F2, F3, F5, F6. F1, F5 and F6 are also reported alone	test run
H2	The scalar index retains at least 80% of the displacement's incremental value	$\bar{Z}$ = mean over test runs of Z_r = ΔR^2_r(c_e) − 0.80 · ΔR^2_r(Disp^{D*}). Supported if the lower bound of $\bar{Z}$ is ≥ 0; equivalent (retention indistinguishable from 80%) if the interval lies within ±0.01. Tested only if G2b passes. The ratio of means and its Fieller interval are reported descriptively	test run
H4a	c_e helps a covariate-aware rehearsal predictor	Mean paired difference in T3, S-pred(B_strong + c_e) − S-pred(B_strong); SESOI δ_B	Stage B test run
H4b	c_e suffices as a standalone rehearsal trigger	Mean paired difference in T3, S-c_e − S-C*; SESOI δ_B	Stage B test run

10.3 Optional exploratory module H3 (path dependence)

H3 is **not** part of the confirmatory study. Its design is preregistered here so that it cannot be tuned after the fact, but it is run only if H1 is supported and the compute budget of §12a remains after Stages A and B (it needs about 120 runs, comparable to the whole Stage A test set). If run, it is reported as exploratory.

The v0.1 design (segments X and Y with high and low key overlap with k) is withdrawn: a global index cannot encode *which* task a stretch of change interfered with, and recency can reverse the sign of the order effect. The redesign varies a **global** property:

- **Blocks.** From one saved state $S(t_0)$ after task k is learned (θ, m, v, data-generator state), two branches are forked. Segment A trains a new task with the same input-token overlap with k as segment B; segment B re-exposes already-mastered data. A two-stage learning-rate schedule (η_high, then η_low) is fixed. Branch 1: A at η_high, then B at η_low. Branch 2: B at η_high, then A at η_low. Steps, tokens, batches, the multiset of examples, Σ η, and C7a are identical across branches.
- **Statement.** Across matched pairs, the order-induced difference in forgetting of k, ΔF, has a positive Spearman correlation with the order-induced difference Δ(Δc_e^{D*}).
- **Decision (descriptive labels).** Supported if the lower bound of the 95% bootstrap interval (resampling unit: the matched pair) is > 0.20; equivalent if the interval lies within (−0.20, 0.20); not part of the confirmatory family.
- **Comparator.** The same correlation is reported for C5b (Σ η‖g̃‖), which is also order-sensitive. H3 shows that c_e tracks a path effect invisible to order-invariant counts; it does not, by itself, show an advantage over order-sensitive telemetry.

10.4 Exploratory, labelled as such

D comparisons beyond D*; the SGD arm; T1-acc and T2; C7b, C8, C8b and C9 bounds; single-competitor comparisons; B_standard comparisons; scale-drift robustness checks; per-family results other than those named above.

11. Gates

Gate	Check	Criterion	Consequence of failure
G1, ceiling	Within-run correlation r_AB between F_A and F_B on calibration runs, uncorrected	$\bar{r}_{AB} \geq 0.70$	Increase probe-set size before freeze; report all skills also as a fraction of $\bar{r}_{AB}$
G2, identifiability	Out-of-sample R^2 for predicting the interval rate $\bar{\rho} = \Delta c_e^{D*} / \max(\Delta N, 1)$ from B_strong, with OLS and with GBDT (fitted on training runs, evaluated on test triplets of F2, F3, F5, F6 and of F5 alone)	Both $R^2 < 0.90$	H1 is reported as *not identifiable*, never as equivalence
G2b, H2 denominator	Mean ΔR^2 of $Disp^{D*}$ over B_strong	Lower bound of its 95% interval > 0.02	H2 is reported as *not identifiable*
G3, competence and headroom	On calibration runs (verified on validation runs): B_strong beats X_cov alone in F1; headroom = $\bar{r}_{AB} - \bar{R}^2$(B_strong) on F2–F3	$\Delta R^2 \geq 0.10$; headroom $\geq 2.5\,\delta$	Redesign the target or the stream before freeze
G3b, baseline saturation	On calibration runs: B_strong re-tuned with a doubled search (100 configurations instead of 50), same seeds for the first 50	Change in mean R^2(B_strong) ≤ 0.01	The baseline is under-tuned: increase the search budget for every predictor set before freeze; if not achievable, H1 may not be reported as supported
G4b, positive control (sensitivity)	On calibration runs only: a synthetic predictor Z carrying known incremental information about T1 is added to B_strong. Z is the out-of-fold residual of T1 given B_strong, mixed with independent noise so that its true ΔR^2 is 2δ. The full pipeline (weighting, locked hyperparameters, GBDT, bootstrap, decision rule) is applied	Estimated ΔR^2 within ± 0.01 of 2δ, and outcome *supported*	The pipeline cannot detect an effect of the size sought: fix the pipeline before freeze. If it still fails, an *equivalent* outcome of H1 may not be reported as closing Level 2
G4, leakage and placebo	R ∩ probes = ∅ and P_ref ∩ P_target = ∅ (asserted in code); incremental value over B_strong, with locked hyperparameters, of a **within-run block-permuted** c_e: the run's per-step increments are cut into 200-step blocks (idle zeros kept in place), the blocks are permuted within the run, and the index is re-accumulated. Evaluated within each family	Point estimate ≤ 0.005 and upper bound of the one-sided 95% interval < 0.015	Stop; leakage or a design artefact
G5, local statistics	$\bar{R}^2$(X_cov + c_e) versus $\bar{R}^2$(X_cov + C6) on test runs	c_e must exceed C6	Otherwise c_e is reported as failing admissibility criterion (a), whatever H1 shows
G6, onset versus skill	H1 statistic with the number of training runs halved and doubled; Stage B predictors with their training budget doubled	No flip between *supported* and *equivalent*, and |change in mean ΔR^2| ≤ 0.01	Report as onset-dependent
G7, reproducibility and budget	Bit-identical reruns of a random 5% of runs; configuration hashes verified before analysis; realised rehearsal steps = Q on every Stage B run	100%	No analysis until fixed

Gate	Check	Criterion	Consequence of failure
G8, reliance	Placebo c_e stream in Stage B (§9)	Upper bound of the one-sided 95% interval of the placebo scheduler's advantage over its comparator < δ_B / 2	c_e is decodable but not used
G9, H3 identifiability	Split-half reliability (A vs B) of ΔF across matched pairs; cross-seed reliability of Δ(Δc_e^{D*}); its coefficient of variation SD / mean Δc_e^{D*}	≥ 0.50; ≥ 0.50; ≥ 0.05	H3 is reported as *not identifiable*

12. Data splits and sample size

- **Calibration runs:** used only for G1, G3, the SESOI value, the choice of D*, m_ch, the stationarity check of R, the variance of the test statistics, and threshold design. They are discarded afterwards and never enter test statistics.
- **Training runs:** F1–F3. **Validation runs:** F1–F3, new seeds. **Test runs:** F1–F3, new seeds, plus F4 (sanity), F5, F6, and the O blocks. O blocks are used only for H3 and G9.
- **Fixed sizes (indicative; frozen after the power simulation):** calibration 24 runs (8 per F1–F3); training 30 base (15 and 60 for G6); validation 15; O-block pairs 60; Stage B 40 paired seeds per scheduler.
- **Sample size for H1.** Let $\hat{\sigma}_{up}$ be the upper 80% chi-square confidence bound of the SD of per-run ΔR^2 on calibration runs. For TOST at a true ΔR^2 of 0 and for the MET at a true ΔR^2 of 2δ, each at power 0.80 and one-sided $\alpha = 0.025$:
  - $n_{TOST} = \lceil (z_{0.975} + z_{0.90})^2 (\hat{\sigma}_{up}/\delta)^2 \rceil = \lceil 10.51 (\hat{\sigma}_{up}/\delta)^2 \rceil$;
  - $n_{MET} = \lceil (z_{0.975} + z_{0.80})^2 (\hat{\sigma}_{up}/\delta)^2 \rceil = \lceil 7.85 (\hat{\sigma}_{up}/\delta)^2 \rceil$;
  - $n = \max(n_{TOST}, n_{MET})$, with at least $\lceil 7.85 (\hat{\sigma}_{up}/\delta)^2 \rceil$ runs each in F5 and in F6 so that their standalone reports have power 0.80 for the MET.
- **Sample size for H2, H4a, H4b** uses the first Holm level, one-sided $\alpha = 0.025/3$ ($z = 2.394$): $n = \lceil (2.394 + z_{0.90})^2 (\hat{\sigma}_{up}/\delta)^2 \rceil = \lceil 13.5 (\hat{\sigma}_{up}/\delta)^2 \rceil$, with δ_B and the Stage B variance for H4.
- **Power simulation.** Before freeze, these closed-form sizes are checked by simulation on calibration data, with the weighted R^2, the final bootstrap, the fixed-sequence and Holm procedure, and the separate F5 and F6 reports.
- **Budget cap:** 200 test runs in Stage A. If n exceeds the cap, the scope is reduced (fewer families, declared before freeze). The SESOI rule is not relaxed.

12a. Stage 0: calibration and go/no-go decision

Stage 0 uses 8–12 technical runs, followed by the calibration runs of §12. It measures and decides; it tests no hypothesis. The author decides after Stage 0 whether to continue. Continuation requires all of the following; otherwise the design is simplified again or the study stops, and the decision is reported.

Criterion	Requirement
Target measurable	G1 passed ($\bar{r}_{AB} \geq 0.70$); at least 20 valid exposure episodes per run; forgetting with non-negligible variance
Index not reconstructible	G2 passed on calibration runs (interval rate, OLS and GBDT). The raw increment Δc_e is not used for this check, because it is dominated by interval length
Headroom	G3 passed (headroom ≥ 2.5 δ)
Baseline and pipeline	G3b passed (baseline saturated); G4b passed (positive control detected)

Criterion	Requirement
Cost	Measured time per full run (training, probes, D3, snapshots) ≤ 40 min single-threaded; projected Stage A ≤ 150 CPU-hours; runs resumable from checkpoints; parallel runs bit-identical (G7)
Sample size	Projected n for H1 ≤ 150 test runs

The feasibility of the full design is not assumed: the step times in §14 cover training only, and probe evaluation and D3 are measured in Stage 0.

13. Stage C: transfer to pretrained language models

Stage C keeps the **question and the inferential logic** (B_strong versus B_strong + c_e, three-way outcomes, gates). It does **not** keep every operational definition, and it is **not part of this preregistration**. A separate Stage C adaptation document will be frozen and deposited with its own DOI before any Stage C run. Its cost is not included in §14. Changes already known:

- **Agent.** Publicly available pretrained causal LMs (the Pythia family); a CPU feasibility check on the smallest model; the confirmatory model size and optimizer arm are named in the adaptation document before any Stage C run.
- **No childhood phase.** Pretraining replaces it; R is a fixed generic-text battery evaluated, not trained.
- **Targets.** T1 is the loss on held-out samples of earlier domains, read at fixed target positions. The accuracy-based inclusion criterion is replaced by a loss-based learned criterion defined in the adaptation document.
- **Covariates.** Task type is replaced by domain identity; competence uses a held-out half of each domain's samples; C7a only (no oracle labels).
- **LoRA.** D is computed on the effective weight update $\Delta W_{eff} = (\alpha/r)(B_{\tau+1}A_{\tau+1} - B_\tau A_\tau)$, not on the concatenated adapter matrices.
- **D3.** Restricted to a small number of target positions and to top-K or sketched distributions; the vocabulary of ≈ 50,000 tokens makes full distributions infeasible.
- **Displacement and C8b, C9.** Evaluated only for the smallest model, for LoRA, or for a subset of tasks, as offline references.

The Stage A conclusions are claimed to generalise only if Stage C reproduces the three-way outcome of H1.

14. Compute estimate

Measured at drafting on the author's laptop (16 logical CPUs), with the §3 agent (0.93 M parameters, untied embeddings, AdamW, clipping):

Setting	ms per training step
batch 32 × 4 tokens, 1 thread	19.9
batch 32 × 4 tokens, 8 threads	17.3
batch 32 × 64 tokens, 1 thread	206.5
batch 32 × 64 tokens, 8 threads	87.5

Because examples are 4–6 tokens long and unpacked (§3), the relevant figure is ≈ 20 ms per step on one thread, and eight single-threaded runs proceed in parallel with almost no loss.

Item	Estimate (to be replaced by Stage 0 measurements)
Training, ≈ 23,000 steps	≈ 8 min per run
Probe evaluation (≈ 120 evaluations × up to 40 tasks × 256 items, forward only)	≈ 1–2 min per run
D3 every 10 steps (2,300 forward passes on 256 items)	≈ 1–2 min per run
One run	≈ 10–12 min single-threaded
Stage A (≤ 200 test runs + ≈ 70 calibration, training and validation runs)	≈ 50 CPU-hours, ≈ 7 wall-clock hours on 8 workers
Stage B	≈ 20–30 CPU-hours
Stage C-mini (smallest Pythia model, CPU)	hours per run; feasibility only

15. Interpretation table (frozen with the protocol)

Outcome	Interpretation
H1 supported, G2 and G5 passed, and F5 supported alone	Accumulated own change predicts own forgetting beyond clocks, counts, and a strong set of logged summaries, and the result generalises across generator parameters: first empirical support for the experiential index (Level 2)
H1 supported, but F5 not supported alone	The advantage does not generalise beyond the training generators; reported as a construction effect, not a general result
H1 supported, and F6 alone shows c_e worse than B_strong (upper bound < 0)	The index is useful where change and interference coincide, and misleading where they do not; reported as a conditional result
H1 equivalent, and G3b and G4b passed	Counts and logged summaries suffice; Level 2 is closed for this regime
H1 equivalent, but G4b failed	Not interpretable as closure: the pipeline could not detect an effect of the size sought
c_e beats B_standard but H1 is equivalent	The information in c_e is reconstructible from cheap telemetry
G2 fails	Not identifiable in this design
G5 fails	c_e fails admissibility criterion (a), whatever H1 shows
F1 alone	Equivalence on F1 is the expected result of the positive control and does not weaken H1
H2 supported	The scalar index captures at least 80% of what full displacement captures, at the storage ratio reported for D*
H2 not supported, or G2b fails	Displacement carries information that the path index does not, or neither adds enough to compare; the claim of a compact index is weakened
H3 supported (G9 passed; optional module)	The index tracks a global path effect invisible to order-invariant counts (exploratory)
H4a supported and G8 passed	The agent uses the index inside a covariate-aware predictor to improve a decision about itself
H4b supported and G8 passed	The index alone suffices as a rehearsal trigger, beyond the best cheap trigger chosen on validation
Stage C reproduces H1	The result is not an artefact of the synthetic world

16. Threats to validity and mitigations

Threat	Mitigation
Designer degrees of freedom in the stream families	F1 natural; F5 and F6 held out; F6 adversarial; families frozen; Stage C on real data
Omitted cheap competitor reconstructs c_e	B_strong with LR-weighted, momentum, work, norm, local and canary summaries; G2 on the rate; B_standard reported
Shared noise between predictors, inclusion, and target	Half B for covariates and inclusion, half A for the target; R and P_ref disjoint from probes and P_target; block-permuted placebo G4; T2 not confirmatory
Scale and weight-decay drift	D1_rel; C6b; $N(t_0)$ and state norms in X_cov; T1-acc
Local-statistics shortcut (closed study)	C6 with short clamped windows inside B_strong; G5
Over-weighting of early tasks	Equal weight per exposure episode
Dependence of triplets within a run	Run-level inference; within-run weighted R^2
Selection of D, k or comparators after seeing results	D*, m_ch and S-C* chosen on calibration or validation and frozen
Unequal rehearsal budgets	Hard cap, pacing, terminal flush; G7
Under-tuned baseline produces a spurious advantage	Equal tuning budget; locked hyperparameters; saturation gate G3b
A null result produced by an insensitive pipeline	Positive control G4b
Optimizer specificity	SGD arm (exploratory)
Scale	Stage C; restrained claims

17. Deliverables

- Code (MIT), data and reports (CC BY 4.0), frozen configurations with SHA-256, a run inventory, automated audit checks, and scripts that regenerate every reported table.
- A Zenodo preregistration (with DOI) before main runs, stating explicitly that the study is a change of the plan announced in the Perspective (versions 1.0–1.2).
- A preprint with *References* links to the Perspective and the closed study, and *Is supplemented by* to the code archive.

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References added in v0.2 (Biderman et al., Hu et al., Kornblith et al., Loshchilov and Hutter, Zenke et al.) are to be verified against arXiv before freeze, as are all others.

19. Revision log v0.1 → v0.2

Sources: **A** = Google Argon review of v0.1 (truncated in Q12; a continuation was requested); **C** = Copilot analysis of that review; **O** = the author's own pre-review. Every accepted item was checked before adoption.

Accepted

Change	Sections	Source
Half B for covariates and inclusion; half A for the target; fact tasks enlarged to 256 items; balanced exposure	§4.1, §4.2, §6, §7	A, O
G1 ceiling uses the uncorrected r_AB (the target uses one half)	§7, §11	A, O
Same-step D2 cancels the Adam preconditioner (verified algebraically); D2 uses a segment-lagged $\hat{v}$; C5d added; Disp^{D2} anchored at $\hat{v}(t_0)$	§5, §7	A, O
D1_rel against scale and weight-decay drift; $\Delta\theta$ logged before and after decay	§3, §5	A
D2 defined in the SGD arm by a passive EMA	§3	A, O
Clipped and raw gradient norms both logged and used	§3, §7	A
B_strong with LR-weighted gradient and loss, squared gradient, SI work, momentum norm, noise scale, state norms, $N(t_0)$; B_standard kept descriptively	§7	A (two-tier design: C)
Canary loss C10 on R	§7	O
C6 windows shortened to $W_{eff} = \min(20, \lfloor \Delta N/4 \rfloor)$ and clamped; minimum interval of 50 steps	§6, §7	A
π_overlap removed from X_cov; C7 split into agent-observable C7a and oracle C7b; agent-observable novelty and overlap covariates	§7	A
R and B_beh on a disjoint key pool; stationarity check of D3	§4.1, §5	A
Storage accounting: one running scalar plus one snapshot per task; snapshot objects per D; C9 storage and training-items gradient	§1, §5.2, §7	A
T1-acc secondary target	§6	A
Operational specifications: answer-position reading, base-2 JS, vocabulary partition, rule-task template, no packing, fixed childhood with gate, O blocks forked from one state, fixed C1 basis, end of interval before the next exposure	§3–§7	A
Monotone constraints specified per feature; locked-hyperparameter ΔR^2; ridge transforms and scaling	§8	A
Equal weight per exposure episode	§6, §8	A
Cross-fitting over training runs; two-sample bootstrap for ridge only	§8	A (scope: C)
H2 as the linear contrast Z_r with denominator gate G2b and an equivalence bound	§10, §11	A, O
H3 redesigned around a global property and moved outside the confirmatory family; G9 expanded; threshold 0.20	§10.3, §11	A (secondary status: C)
Stage B: hard budget cap, pacing, terminal flush; comparator chosen on validation; S-uniform, S-C2, S-C5, S-C5b added; H4 split into H4a and H4b; online covariates from training batches	§9, §10	A, O
Within-run block-permuted placebo; G4 one-sided, locked hyperparameters, per family; cross-seed yoked stream kept as a robustness check	§9, §11	A (both kept: C)
Single $\alpha = 0.025$ per side for MET and TOST; fixed sequence H1, then Holm over H2, H4a, H4b; Stage B only after H1 is supported	§9, §10	A, C, O
Power formulas with $z_{1-\beta/2}$ for TOST, the Holm level, an upper bound on σ, and F5/F6 allocation; checked by simulation	§12	A, C
Headroom-based SESOI rule, frozen before test, with raw and relative gains both reported	§10.1	A (caution: C)
G2 on the interval rate with OLS and GBDT; G3 on calibration with a headroom check; G5 and G6 made quantitative; G8 one-sided	§11	A, O

Change	Sections	Source
F4 not pooled; adversarial F6 added; F5 described as generalisation across generator parameters; F1 row added to the interpretation table	§4.4, §15	A, O (F6 design: C)
Stage C rewritten as a transfer protocol with a separate adaptation document; LoRA on ΔW_eff; no childhood	§13	A, C, O
Compute estimate replaced by measured step times; single-threaded workers for determinism and throughput	§3, §14	O, A
Notation: $\bar{\rho}$ for the rate, m_ch for the chunk stride, L_reh for the rehearsal length; §2 wording on elapsed time; consistent scheduler names	throughout	A
Content versus provenance note	§0.4	O
Prior work FOREVER (Feng et al. 2026) cited, with priority for the index acknowledged; contribution restated as an identification audit; exploratory S-FOREVER arm. Verified on arXiv and in the ACL Anthology: $\tau_t = \Sigma\|\Theta_i - \Theta_{i-1}\|_2$ on LoRA weights; only step counts tested as an alternative clock	§0.3, §9, §18	A (named), C (positioning), O (verification)
H3 made an optional exploratory module, run only if H1 is supported and budget remains	§10.3, §15	C
Stage 0 go/no-go criteria (cost, sample size, episodes per run)	§12a	C
Stage C declared outside this preregistration	§13	C
Positive control G4b (sensitivity of the pipeline to an effect of 2δ) and baseline-saturation gate G3b (doubled tuning budget); interpretation table and Stage 0 criteria updated. Motivated by the second priority-search round on the closed study (positive-control validation; Mukhoti et al. 2018)	§11, §12a, §15	C, O
Mukhoti et al. 2018 (strong baselines), Knopov and Shvai 2026 (TOST and run counts for stochastic ML), Elazar et al. 2021 (decodability is not use) cited; all verified on arXiv or Crossref	§7, §9, §10.1, §18	C (priority search on the closed study)

Rejected or modified

Suggestion	Decision
"PyTorch 2.13 and NumPy 2.5 do not exist" (A)	Rejected: both are installed and were used for the measurements (PyTorch 2.13.0, NumPy 2.5.1)
Per-step D2 weighted by the running maximum of $\hat{v}$ (A)	Modified: segment-lagged $\hat{v}$ is used, because a running maximum never decreases and would track history age
Weight-decay pulses and pre-LN rescaling as part of F6 (A)	Moved to exploratory robustness checks: no counterpart in ordinary training (C)
H3 option B (midpoint evaluation or a recency-weighted index) (A)	Rejected: it introduces a second index and complicates interpretation (C)
Packed 64-token sequences with block-diagonal masks (A)	Rejected: examples are unpacked, which is simpler and, as measured, ≈ 4× cheaper per step
Argon's split sizes adopted as final (A)	Modified: indicative only, to be fixed by the power simulation (C)
Two-sample bootstrap with refitting for all models (A)	Modified: ridge only; cross-fitting for GBDT (C)
Go/no-go on $R^2(\Delta c_e \mid B_{strong}) < 0.90$ for the raw increment as well as the rate (C)	Rate only: the raw increment is dominated by interval length and would fail the check trivially
Full-run time of 45–120 min (C)	Not adopted as an estimate: it assumed long sequences; examples are unpacked. Replaced by a measured go/no-go limit of 40 min per run (§12a)

Pending. The truncated part of the Argon review (end of Q12, Q13, compute check (g), §15–§17, and its five priority changes) has been requested. Any further change will be logged in v0.3.

Update of 1 November 2026 (no change of design). The Perspective (v1.3, 10.5281/zenodo.23847648), the study (v1.3, 10.5281/zenodo.23847602), and the code archive (v1.3, 10.5281/zenodo.23847723) were revised after an automated review by the CMU Paper Reviewer. Citations and the closure rule in §0 now follow v1.3: non-inferiority within a prespecified margin and no worse declared cost, with mixed trade-offs reported as unresolved.